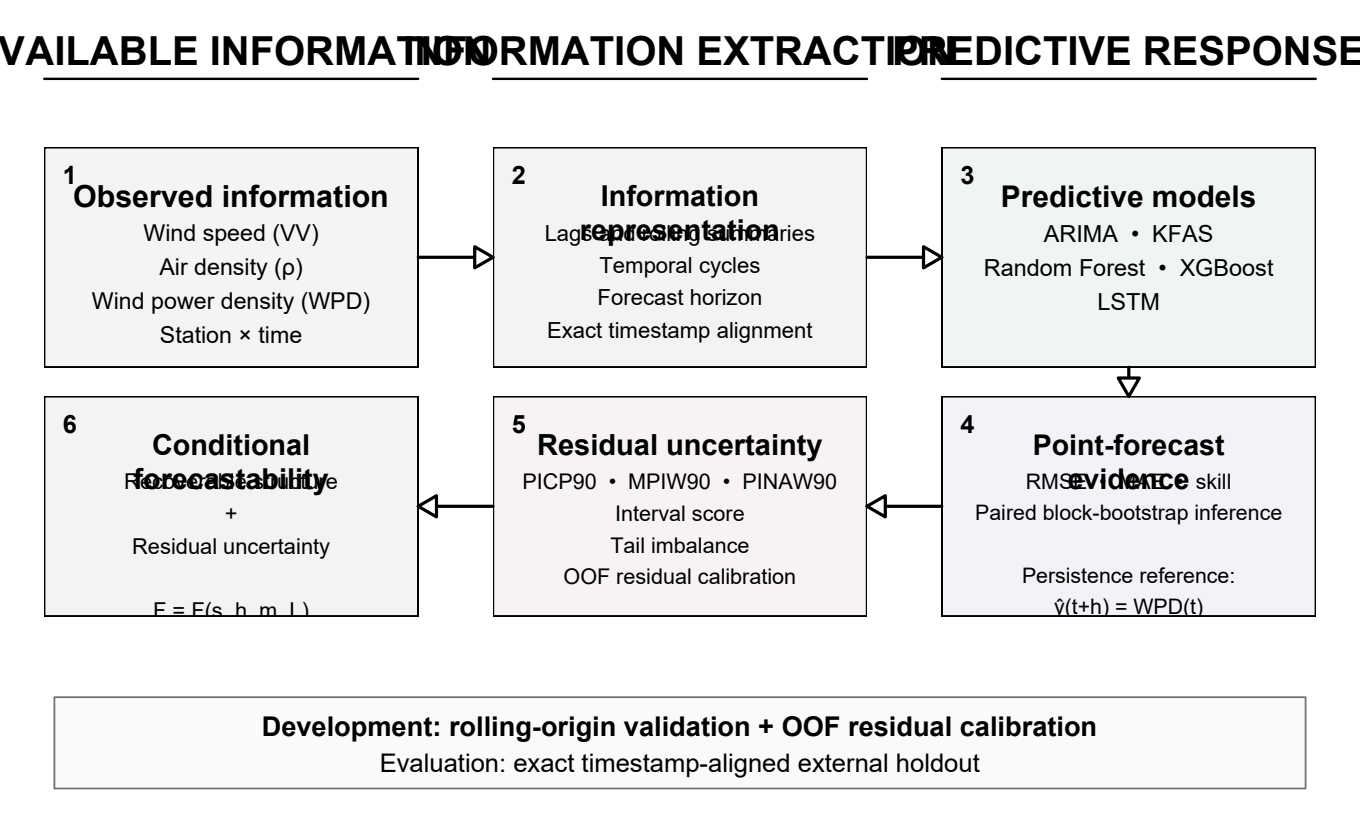
AVAILABLE INFORMATION
INFORMATION EXTRACTION
PREDICTIVE RESPONSE
1
Observed information
Wind speed (VV)
Air density (ρ)
Wind power density (WPD)
Station × time
2
Information representation
Lagged and rolling summaries
Temporal cycles
Forecast horizon
Exact timestamp alignment
3
Predictive models
ARIMA • KFAS
Random Forest • XGBoost
LSTM
4
Point-forecast evidence
RMSE • MAE • skill
Paired block-bootstrap inference
Persistence reference:
ŷ(t+h) = WPD(t)
5
Residual uncertainty
PICP90 • MPIW90 • PINAW90
Interval score
Tail imbalance
OOF residual calibration
6
Conditional forecastability
Point-forecast evidence
+
Residual uncertainty
F = F(s, h, m, I)
Development: rolling-origin validation + OOF residual calibration
Evaluation: exact timestamp-aligned external holdout